Switch Case Statements :

Question 1 :

A user is given several options as part of a menu to calculate the surface area of different shapes. Write a program to input a number as an option for a menu, and execute the program accordingly.

The user is given the following options:

1. Find the surface area of a cylinder ($2\pi r^2 + 2\pi rh$)
2. Find the surface area of a sphere ($4\pi r^2$)
3. Find the surface area of a cone ($\pi r^2 + \pi rl$)
4. Find the surface area of a cube ($6s^2$)

Take the necessary values as input from the user, calculate the surface area of the shape, and display it.

Question 1 :

A user is given options in a menu to calculate the simple interest or the compound interest. Take the option selected as input and display the correct value accordingly.

The user is given the following options:

a. Find the Simple interest for a principal amount.

$$\frac{Interest = Principal \times time \times rate}{100}$$

b. Find the Compound Interest for a principal amount

$$Interest = principal \times \left[1 + \frac{r}{100}\right]^n - 1$$

Question 3 :

A user is given multiple options to perform a mathematical function on a decimal value. Take the option and value as input, perform the necessary mathematical operation and display the answer.

The user is given the following options.

1. Find the square root of the value
2. Find the cube root of the value.
3. Round the value
4. Round the value to the next integer.
5. Round the value to the previous integer.

Question 4 :

The user is given options to find the maturity value or the interest on a recurring deposit account. Take the option as input from the user, calculate the necessary value and display it.

The user is given the following options.

1. Find the maturity value on a sum.
   $$M.V. = p \times n + p \times n \times \frac{(n+1)}{24} \times \frac{r}{100}$$
2. Find the interest on a sum
   $$Interest = p \times n \times \frac{(n+1)}{24} \times \frac{r}{100}$$